

Probability Lecture 6

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Variance

A measure of spread of a PMF

- Random Variable X , with mean $\mu = E[X]$
- Distance from the mean: $X - \mu$
- Average Distance from the mean?
 $E[X - \mu] = E[X] - \mu = \mu - \mu = 0 \dots$ not useful

Definition of Variance : $\text{var}(X) = E[(X - \mu)^2]$

$\text{var}(X) = E[g(x)]$ where $g(x) = (x - \mu)^2 = \sum_x (x - \mu)^2 p_X(x)$

Standard deviation: $\sqrt{\text{var}(X)}$

Properties of the Variance

$$\text{var}(aX + b) = a^2 \text{var}(X)$$

Let $Y = X + b$

$$\nu = E[Y] = \mu + b$$

$$\text{var}(Y) = E[(Y - \nu)^2] = E[(x + b - (\mu + b))^2] = E[(x - \mu)^2] = \text{var}(X)$$

Let $Y = aX$

$$\nu = E[Y] = a\mu$$

$$\text{var}(Y) = E[(aX - a\mu)^2] = E[a^2(x - \mu)^2] = a^2 E[(x - \mu)^2] = a^2 \text{var}(X)$$

Another Formula

$$\text{var}(X) = E[X^2] - (E[X])^2$$

We have

$$\text{var}(X) = E[(x - \mu)^2] = E[X^2 - 2\mu X + \mu^2] = E[X^2] - 2\mu E[X] + \mu^2$$

but $\mu = E[X]$

$$\Rightarrow E[X^2] - (E[X])^2$$

Variance of the Bernoulli

i.e. $X = 1, 0$ with probability $p, 1 - p$

$$\text{var}(X) = \sum_x (x - E[X])^2 p_X(x) = (1 - p)^2 p + (0 - p)^2 (1 - p) = p(1 - p)$$

or

$$(var)(X) = E[X^2] - (E[X])^2 = p - p^2 = p(1 - p)$$

as $X^2 = X$ for a Bernoulli PMF

Variance of the uniform

$$\text{var}(X) = E[X^2] - (E[X])^2 = \frac{1}{n-1}(0^2 + 1^2 + 2^2 + \dots + n^2) - \left(\frac{n}{2}\right)^2 = \frac{1}{12}n(n+2)$$

more generally sub in $b - a = n$

Note

Could add more formulae to calculate $\text{var}(X)$

Conditional PMF and expectation, given an event

Condition on an event $A \Rightarrow$ use conditional probabilities

$$p_X(x|A) = P(X = x|A)$$

$$\sum_x p_{X|A}(x) = 1$$

$$E[X|A] = \sum_x x p_{X|A}(x)$$

$$E[g(X)|A] = \sum_x g(x) p_{X|A}(x)$$

$$\text{var}(X) = E[(X|A - E[X|A])^2]$$

Total Expectation Theorem

We know the total probability theorem

$$P(B) = P(A_1)P(B|A_1) + \dots + P(A_n)P(B|A_n)$$

where $A_1 \dots A_n$ form a partition of the sample space

Then, for all x

$$P_X(x) = P(A_1)p_{X|A_1}(x) + \dots + P(A_n)p_{X|A_n}(x)$$

$$\sum_x x P_X(x) = P(A_1) \sum_x x p_{X|A_1}(x) + \dots$$

Then the total expectation theorem is

$$E[X] = P(A_1)E[X|A_1] + \dots + P(A_n)E[X|A_n]$$

Conditioning a geometric random variable

X : number of independent coin tosses until first head; $P(H) = p$

$$p_X(k) = (1-p)^{k-1}p, \quad k = 1, 2, \dots$$

Memorylessness: Number of remaining coin tosses conditioned on Tails in the first toss is geometric with parameter p

i.e. Conditioned on $X > n$, $X - n$ is geometric with parameter p

$$P_{X-1|X>1}(3) = P(X-1=3|X>1) = P(T_2T_3H_4|T_1) = P(T_2T_3H_4) = (1-p)2p = P_X(3)$$

$$P_{X-n|X>n}(k) = p_X(k)$$

The mean of the geometric

From the definition

$$E[X] = \sum_{k=1}^{\infty} k p_X(k) = \sum_{k=1}^{\infty} k (1-p)^{k-1} p$$

Or,

$$E[X] = 1 + E[X-1] = 1 + pE[X-1|X=1] + (1-p)E[X-1|X>1] = 1 + 0 + (1-p)E[X]$$

$$E[X] = \frac{1}{p}$$

Multiple Random Variables and joint PMFs

$$X : p_X, \quad Y : p_Y \quad (\text{marginal PMFs})$$

Joint PMF: $P_{X,Y}(x,y) = P(X=x \text{ and } Y=y)$

$$\sum_x \sum_y p_{X,Y}(x,y) = 1$$

$$p_X(x) = \sum_y p_{X,Y}(x,y)$$

$$p_Y(y) = \sum_x p_{X,Y}(x,y)$$

More than two random variables

$$p_{X,Y,Z}(x,y,z)P(X=x, Y=y, Z=z)$$

$$\sum_x \sum_y \sum_z p_{X,Y,Z}(x,y,z) = 1$$

$$p_X(x) = \sum_y \sum_z p_{X,Y,Z}(x,y,z)$$

$$p_{X,Y}(x,y) = \sum_z p_{X,Y,Z}(x,y,z)$$

Functions of multiple random variables

$$Z = g(X, Y)$$

$$\text{PMF: } P_Z(z) = P(Z=z) = P(g(X,Y)=z) = \sum_{(x,y):g(x,y)=z} p_{X,Y}(x,y)$$

Expected Value Rule

$$E[g(X,Y)] = \sum_x \sum_y g(x,y) p_{X,Y}(x,y)$$

Linearity of Expectation

$$E[aX + b] = aE[X] + b$$

For multiple random variables

$$E[X + Y] = E[X] + E[Y]$$

$$\begin{aligned} E[X + Y] &= E(g(X, Y)), \quad (g(x, y) = x + y) \\ &= \sum_x \sum_y (x + y) p_{X,Y}(x, y) = \sum_x \sum_y x p_{X,Y}(x, y) + \sum_x \sum_y y p_{X,Y}(x, y) \\ &= \sum_x x \sum_y p_{X,Y}(x, y) + \cdots = \sum_x x p_X(x) + \sum_y y p_Y(y) = E[X] + E[Y] \end{aligned}$$

One can easily see that this formula holds for more random variables.

The mean of the binomial

X : binomial with parameters n, p - number of successes in n independent trials

$$E[X] = \sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k}$$

To compute this,

$X_i = 1$ if i th trial is a success, else $X_i = 0$

$$X = X_1 + \cdots + X_n$$

$$E[X] = E[X_1] + \cdots + E[X_n] = np$$